

Agenda

- ① Introduction
- ② Basic Terminologies
- ③ Set Operations
- ④ Addition Rule
- ⑤ Code: Sachin's ODI Performance

Introduction

Assignment

- ① 5 Assignment (mandatory)
- ② 3 H/w Question

PSP : 85% + 95%

I will share additional
Practice Question

Pre-read and Post Read

Case: Aero-fil

① Don't Focus or Try to
memorize formula's
Building Intuition

Basic Terminologies

Experiment

An Activity with some data

① Deterministic Experiment

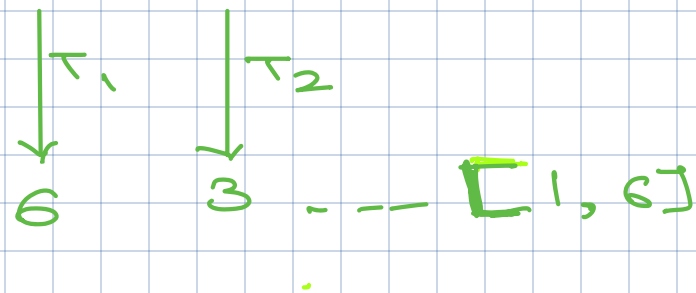
$$a^2 + b^2 + 2ab \Rightarrow 49$$

$$\begin{array}{l} a = 3 \\ b = 4 \end{array}$$


The outcome stays constant
No matter how many times
Exp is repeated

② Probabilistic Exp

① Rolling a die



Terminologies in Probabilistic Exp:

① Outcome: Result

$\{1, 3\}$ $\{2, 3\}$ $\{3, 3\}$ $\{4, 3\}$ $\{5, 3\}$ $\{6, 3\}$

② Sample Space: Collection of all possible outcomes

$\{1, 2, 3, 4, 5, 6\}$

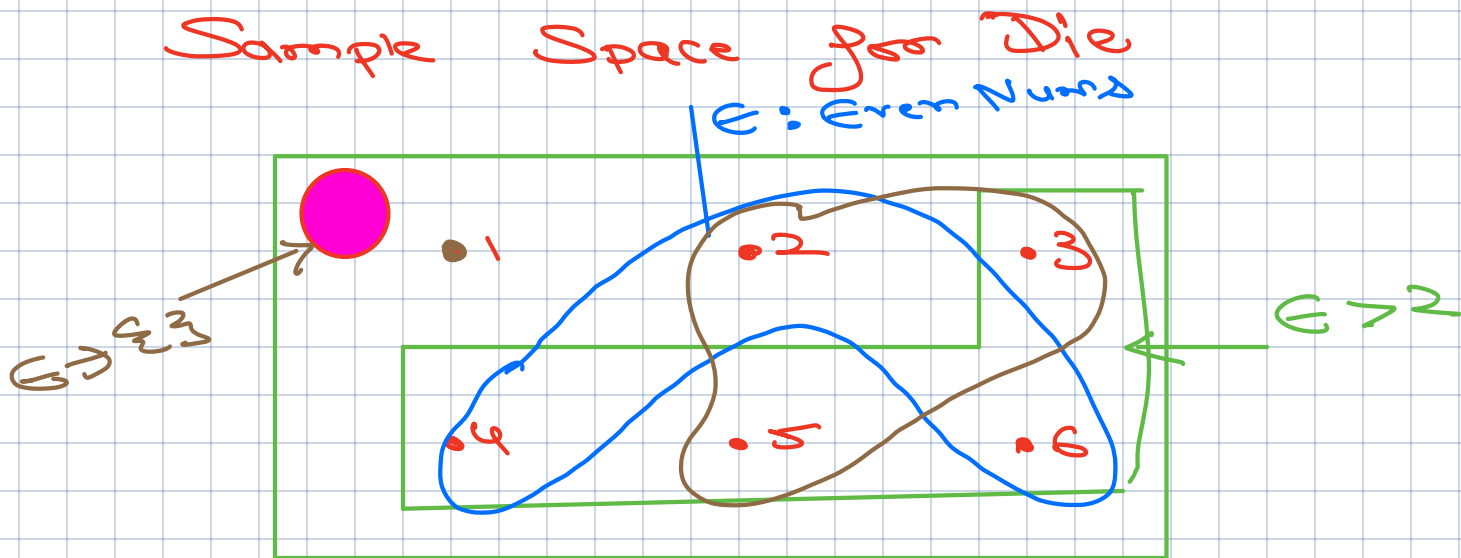
③ Event: Any subset of Sample Space

Event : Getting Even Numbers
as outcome

$$E \ni \{2, 4, 6\}$$

* Since empty set $\{\}$
is a subset of every
set, It is considered a
valid Event

$$P(\{\}) = 0$$

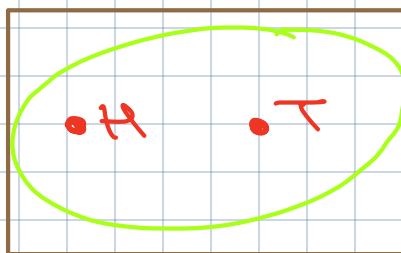


• Outcome Set of Sample Space

$\emptyset$ Event

$\rightarrow$ outcomes $\rightarrow \mathcal{Q}$ Events

Tossing Coin



Event $\mathcal{E}$

Outcomes Ω

$\Omega = \{H, T\}$

What is the P that you will get either H or T

$\mathbb{P}$ $\begin{cases} \{H\} & \text{for dice roll} \checkmark \\ \{H, T\} & \text{for coin toss} \times \end{cases}$

Key Takeaway: An Event must always be a subset of sample spaces

i.e

Outcomes in Event should be present in Sample Space

$$\text{Probability of Event} = \frac{\# \text{ Outcomes in Event}}{\text{Total Outcomes}}$$

$$= \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

$$P(H) = \frac{1}{2} \quad \Sigma H$$

$$P(T) = \frac{1}{2} \quad \Sigma T$$

$$P(H \text{ or } T) = \frac{2}{2} \quad \Sigma H, T$$

$$P(\quad) = \frac{0}{2} = 0 \quad \Sigma \quad$$

$$P(\text{Even Number}) = \frac{3}{6} = 50\%$$

~~H~~ 5

Σ 1, 3, 5, 7

Not a possible outcome

2 x 6 = 12 outcomes

• H1	• H2	• H3	• H4	• H5	• H6
• T1	• T2	• T3	• T4	• T5	• T6

Sample Space for
Coin Toss followed by Dice Roll

Q. We are tossing a coin followed by a dice.
How many possible outcomes will be there in the sample space?

Event $D > 4$ ⇒ H4 H6
T4 T6

Event $\mathcal{D} = 4$ and T $T4$

Event $\mathcal{Q}^{12}$

Set Ops

Dice $\Rightarrow \{1, 2, 3, 4, 5, 6\}$

Sample Space

Universal Set

Shashank (odd Number)

$E-S \Rightarrow \{1, 3, 5\}$

$P(E-S) = \frac{3}{6} = 50\%$

Ashwath (1, 5, 6)

$E-A \Rightarrow \{1, 5, 6\}$

$P(E-A) = \frac{3}{6} = 0.5$

Mohit (Even Numbers)

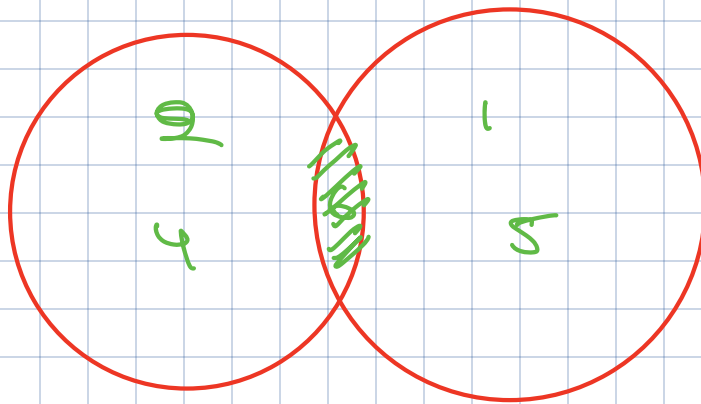
$E-M \Rightarrow \{2, 4, 6\}$

$\frac{3}{6} = 0.5$

11) In which condition both
M-Rith and Aawath will
win

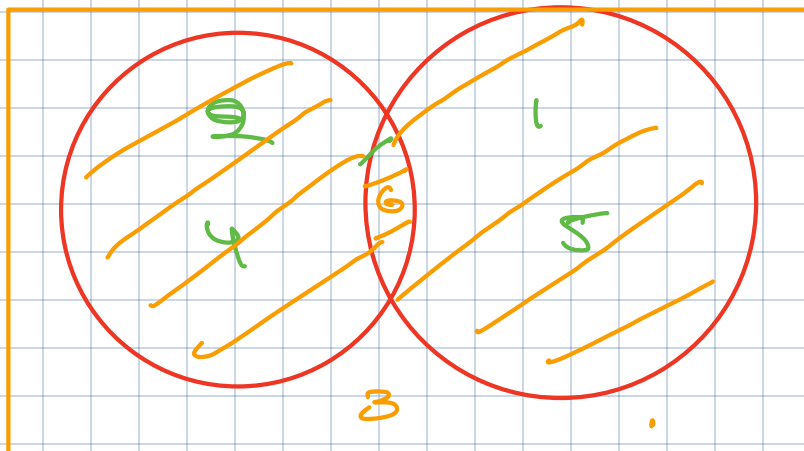
E-M 3 (2 4 6) E-A 3 (1 5 6)

$P(E-M \text{ and } A) = \frac{1}{6}$



Intersection

2) When either M-Rith or
Aawath win 3 {2 4 6, 1 5}



$\frac{5}{6}$

10

When Morit will be the best

$$E_M \supset \{2, 4, 6\}$$

$$E_{-M} \supset \{1, 3, 5\} \text{ or } E_{-M^c} \supset \{1, 3, 5\}$$

Mutually Exclusive Events
Disjoint Events

$$A \cap B \supset \{3\}$$

we can say that A and B
are mutually exclusive/disjoint

$$M, S \quad \checkmark$$

$$A, S \quad \times$$

Exhaustive Event

If union of two or more
Events is equal to Sample
Space

Morit, SRachant $\checkmark$

$$M, S, A \quad \checkmark$$

$$S, A \quad \times$$

$$M, A \quad \times$$

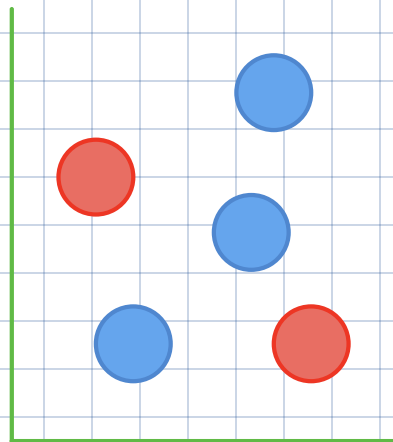
$$A \cup S \cup M \supset \{1, 2, 3, 4, 5, 6\}$$

Independent Events

Exp 1
↓
 E_1

and Exp 2
 E_2

- ① Outcome of Exp 2 $\rightarrow E_2$
Does not Depend on E_1
Then E_1 and E_2 are
independent



E_1 Pick a ball
Check its outcome
put it Back

E_2 Pick another Ball

Independent
Event

E_1 Pick a ball
Check its outcome

E_2 Pick another Ball

Not Independent

110

Q. There are 4 green balls, 6 yellow balls, and 2 blue balls in a bag.
A random ball is chosen.
Find the probability that a yellow or blue ball is chosen.

$$E(Y \text{ or } B) = 6 + 2$$

$$\frac{\# \text{ Outcomes } E}{\text{Total Out}} = \frac{8}{12}$$

Q. Which of the following represent mutually exclusive sets?

4 options

Active Duration (Most preferred: 30 seconds)

Appears for 60 Secs

A Youtube premium Vs Non-premium users



Σ 3

B People who like cappacino vs Espresso

X

C Users of Swiggy vs Zomato

X

D Users of Amazon vs Flipkart

X

A and B

When Both of them can't happen simultaneously

$$C \cap E \neq \emptyset \quad \Sigma 3$$

$$A \cap B = \emptyset \quad \Sigma 3$$

$$C \cap E = \emptyset \quad \Sigma \text{ Protected}$$

Q. It is known that 60% people use Swiggy, 50% use Zomato. 20% people use both.

What percentage use Swiggy, but do not use Zomato?



II : % only use Zomato
 $50 - 20 = 30\%$

II : % only use Swiggy
 $60 - 20 = 40\%$

III : Either Z or S $40 + 30 + 20$

IV : Neither Z or S 10%

Q. Which of the following represent mutually exclusive sets?

2 users have participated



A Youtube premium Vs Non-premium users 100%

B People who like cappacino vs Espresso 0%

C Users of Swiggy vs Zomato 0%

D Users of Amazon vs Flipkart 0%

[End Quiz Now](#)

① T1LOPS

② P and Stats

③ Math for ML and Supervised ML

④ Python

